UNIT I · CHEAT SHEET

Foundations of unmanned aerial systems

UAV vs UAS, history, classification, systems engineering, subsystems, autonomy and safety

UAV vs UAS

- **UAV:** the air vehicle only.
- **UAS:** air vehicle + ground control station + data links + payloads + launch and recovery + ground support + crew + procedures.
- Six core subsystems: air vehicle, MPCS/GCS, payloads, data links, launch and recovery (LRE), ground support (GSE).

History in ten names

- 1917 Hewitt-Sperry Automatic Airplane; 1918 Kettering Bug: preset aerial torpedoes.
- 1935 DH.82B Queen Bee; 1940 Radioplane OQ-2: radio-controlled target drones.
- 1960s–70s Ryan Firebee / Model 147 Lightning Bug: jet reconnaissance.
- 1982 IAI Scout, Tadiran Mastiff (Bekaa Valley); RQ-2 Pioneer; Aquila: a lesson in requirement creep.
- MQ-1 Predator, MQ-9 Reaper (MALE); RQ-4 Global Hawk (HALE).
- Multirotors were enabled by MEMS inertial sensors, LiPo batteries and brushless DC motors.

Four ways to classify

- **Lift:** fixed-wing, rotary-wing, flapping-wing, hybrid VTOL.
- **Mass (MTOM):** < 250 g; 250 g–2 kg; 2–25 kg; 25–150 kg; > 150 kg. Impact energy $E = \frac{1}{2}mv^2$ drives the rules.
- **Range/altitude:** close < 10 km; short 10–50 km; medium 50–200 km; MALE 5–10 km altitude; HALE > 15 km.
- **Airspace:** VLOS (direct sight, about 300–500 m), EVLOS (visual observers), BVLOS (instruments only).

Data links

- **C2 uplink:** < 50 kbps, < 50 ms; flight-critical; loss triggers failsafe.
- **Telemetry downlink:** about 100 kbps; safety-critical.
- **Payload downlink:** 5–50 Mbps; mission-essential; loss does not crash the aircraft.

Models to prototypes

- Conceptual → mathematical → simulation (SIL/HIL) → scale / wind-tunnel → functional prototype → technology demonstrator.
- Model aircraft fly open-loop under hobby rules; UAV prototypes fly closed-loop with failsafes and verifiable requirements.

Systems-engineering V-model

- **Left:** stakeholder needs → system spec → subsystem allocation (mass, power, bandwidth) → component design.
- **Bottom:** build and code.
- **Right:** unit tests → subsystem integration → HIL → flight acceptance against the original needs.

Write a verifiable requirement

- Bad: "fly for a long time".
- Good: "shall cruise at 16.0 ± 1.0 m/s for ≥ 45 min at MSL with a 500 g payload, landing with ≥ 15% reserve".
- Each requirement names its verification: analysis, inspection, ground test or flight test.

Payload integration checklist

- Mechanical: CG within limits, mounting, vibration isolation.
- Electrical: power budget, voltage rails, EMI/EMC.
- Thermal dissipation; data bus protocol; the interface control document (ICD) records it all.

Autonomy and safety

- **Automation:** fixed, deterministic sequence. **Autonomy:** closed-loop sense–plan–act under uncertainty.
- **Lost link:** after about 3–5 s without C2 heartbeat, climb, return to launch, land.
- **Geofence:** cylinder or polygon the autopilot will not cross.
- **Dissimilar redundancy:** baro + GNSS + rangefinder, so one cause cannot fail all.

Exam traps

- Calling the aircraft alone a "UAS".
- Requirements with no number or no test method.
- Mixing up EVLOS (observers see it) and BVLOS (nobody does).